Lane Detection for a Lightweight Driver-Assistance Prototype

Student Research Proposal • Sadril Hossain Ether • Computer Vision • Image Processing • Deep Learning

Proposed by	Sadril Hossain Ether
Project type	Student-led research / prototype study
Estimated duration	10–12 weeks

Project Overview

I want to explore a lightweight computer-vision pipeline for detecting lane boundaries in road images or video. The project will focus on understanding image preprocessing, lane-feature extraction, and the trade-off between accuracy and real-time performance rather than building a full autonomous-driving system.

Student perspective
I want to keep the first version realistic for a student team: a focused research question, a reproducible experiment, and a prototype that can be discussed with a researcher and improved through collaboration.

Research Problem

Lane markings can be affected by shadows, worn paint, curves, weather, and changing camera angles. A student-scale project can provide a useful way to study these challenges while keeping the problem focused on visual lane detection and not on vehicle control or autonomous navigation.

Research Objectives

- Build a baseline lane-detection pipeline for road images.
- Study preprocessing techniques that improve lane visibility.
- Compare a classical image-processing approach with a lightweight learned approach where feasible.
- Measure detection quality and processing speed.
- Analyze failure cases such as shadows, curves, and unclear markings.

Research Questions

RQ1. How well can classical image processing handle simple lane scenes?

RQ2. Which preprocessing steps are most useful for difficult lighting?

RQ3. What is the accuracy-versus-speed trade-off of a lightweight learned model?

RQ4. Which road conditions cause the most detection errors?

Proposed Methodology

Step 1. Select a suitable public road-image dataset.

Step 2. Implement a classical baseline using image transformations and geometric constraints.

Step 3. Prepare a small labeled subset for evaluation or model training if required.

Step 4. Compare with a lightweight learning-based method if resources permit.

Step 5. Evaluate detection quality on different scene conditions and measure processing time.

Step 6. Document failure cases and define realistic limits of the prototype.

Data / Materials / Experimental Resources

Public road-image dataset, Python notebooks, camera/video samples where appropriate, and optional GPU access.

Evaluation Plan

Evaluate lane-boundary overlap or point/line error using a clearly defined metric, together with qualitative inspection and frame-processing time.

Expected Challenges

- Poor lighting and worn markings can confuse the detector.
- Curved roads may violate simple geometric assumptions.
- A small labeled dataset can limit deep-learning performance.
- Real-time performance may require model simplification.

Reproducibility and Documentation

I will keep the main assumptions, dataset versions, preprocessing decisions, experiment settings, and evaluation results documented. The aim is that another student can understand how the result was produced and repeat the core experiment without relying on undocumented manual steps.

Planned Mitigation

I will begin with a small baseline and a feasibility check. If the original scope becomes too large for the available time or resources, I will narrow the research question rather than adding unnecessary complexity. Limitations and failed experiments will be documented as part of the research process.

Expected Outcomes

- A classical lane-detection baseline.
- Optional lightweight deep-learning comparison.
- Accuracy and speed evaluation.
- A structured failure-case analysis and deployment discussion.

Potential Research Collaboration

I am looking for someone interested in computer vision, image processing, or deep learning who can help with dataset preparation, model comparison, or evaluation.

Technical Stack

Python, OpenCV, NumPy, PyTorch or TensorFlow, Jupyter Notebook.

Scope, Ethics, and Responsible Research

The project will use public or appropriately collected road imagery. The scope is limited to lane perception for research; it will not control a vehicle or make safety-critical driving decisions.

Proposed Work Plan

Phase	Activities
1. Background study	Literature review, problem definition, feasibility check
2. Setup	Data/system preparation and baseline implementation
3. Experiments	Development, comparison, tuning, and documentation
4. Evaluation	Metrics, error analysis, limitations, reproducibility
5. Reporting	Results, discussion, documentation, and presentation

What I Am Looking for in a Collaborator

I am looking for someone who is genuinely interested in the research question and willing to contribute to discussion, experiment design, implementation, analysis, or writing. I would like to divide responsibilities according to each person's strengths and keep the work transparent and documented.

Possible Next Step

If a researcher is interested, I would first like to discuss feasibility, refine the research question, and agree on a realistic division of work. The proposal can then be updated into a more specific study plan with milestones.

Note: This is a proposed student project. It does not claim completed research or guaranteed results, and the scope can be refined through collaboration.